

ARCHIT SHARMA

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EDUCATION

New York University

M.S. Mechatronics and Robotics

Sept 2024 - May 2026

GPA: 3.8

- Relevant Labwork: Assisted with Ultra OPI teleoperation data collection to fine-tune a pi0.5 VLA manipulation policy.

Birla Institute of Technology Pilani - Goa

B.Tech Mechanical Engineering

Aug 2019 - May 2023

GPA: 3.4

SKILLS

- **Robotics and Simulation:** ROS/ROS2, NVIDIA Isaac Lab, MuJoCo, RLlib, Stable Baselines 3, MINK/PINK, Docker
- **Machine Learning and Computer Vision:** PyTorch, TensorFlow, OpenCV, Open3D, PCL
- **Programming Languages:** Python, C/C++, C#

PUBLICATIONS

Neural Network-Based Control for Vehicle Suspension Systems with Active Damping

Archit Sharma, Pravin M Singru

The 30th International Congress on Sound and Vibration, 2024

EXPERIENCE

Applied Dynamics and Optimization Laboratory

Graduate Student Researcher

Sept 2025 - Present

NYU, New York

- Integrated COMSOL fluid simulation with MuJoCo to enable energy analysis and trajectory optimization of bio-inspired fish (carangiform) locomotion.
- Developed a convex optimization framework to bridge gaps between COMSOL and MuJoCo and correct known model inaccuracies.

Robotic Design Team

Mechanical Engineer

Jan 2025 - May 2025

NYU, New York

- Designed and prototyped the excavation subsystem for the NYU Lunabotics rover using SolidWorks and Onshape.
- Conducted physical testing of the rover, passing all preliminary tests.

Dynamics and Vibrations Lab

Undergraduate Student Researcher

Aug 2021 - Jan 2024

BITS Pilani, Goa

- Implemented and benchmarked PD, PID, and LQR controllers for a quarter-car semi-active suspension system.
- Researched, implemented, and published a novel neural network-based adaptive controller for the system.

Computational Intelligence Lab

Research Intern

Jan 2023 - Aug 2023

IISc, Bengaluru

- Conducted musculoskeletal analysis of deep skeletal muscles using inverse kinematics (IK) and dynamics (ID) on OptiTrack motion captures for several participants.
- Developed a signal processing pipeline in MATLAB using FFT and signal filters (band-pass, Kalman) to successfully denoise and analyze IMU and EMG sensor data.

PROJECTS

Optimal Control and Reinforcement Learning 🔄

Sept 2025 - Dec 2025

- Trained a Unitree Go2 quadruped robot in NVIDIA Isaac Lab using PPO deep reinforcement learning algorithm with a custom reward model to achieve a stable gait on rough and stepped terrains.
- Implemented a nonlinear MPC enabling simulated drones to execute agile maneuvers.
- Implemented an MPPI controller for a 3R planar manipulator to achieve accurate end-effector trajectory tracking.

Robot Perception and Computer Vision 🔄

Sept 2025 - Dec 2025

- Implemented a custom Optical Flow algorithm in OpenCV for multi-object vehicle tracking.
- Developed a content-based image retrieval pipeline using a ResNet feature extraction model trained on the VPR dataset to accurately identify locations in New York.

Formation Control and Flocking for Drone Swarms

Mar 2025 - May 2025

- Programmed decentralized simulated drone swarms using Reynolds' Flocking theorem to implement collective behavior in Unity under safety constraints enforced through a potential field controller.
- Implemented and tested swarm optimization algorithms such as PSO for exploratory theory.

Gesture Controlled Hexapod

Nov 2024 - May 2025

- Programmed an Arduino microcontroller to drive a hexapod, interfaced with an ESP32 module to stream video feed over Wi-Fi.
- Designed a wrist-mounted ESP32 controller to sample analog inputs from flex sensors and wirelessly transmit control states.
- Implemented gait algorithms assigned to different gestures for stable, multidirectional legged motion.